DEGREE PROJECT

A Strategic Roadmap for Swedish Forestry Contractors

Supporting Adaptation Under Ongoing Industry Transformation

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ABSTRACT

Purpose - The purpose of this study is to develop a strategic roadmap to support Swedish forestry contractors in maintaining competitiveness under changing industry conditions. The study addresses two research questions: how ongoing transformations in the forestry industry impact Swedish forestry contractors, and how a strategic roadmap can be designed to support their adaptation.

Method - A qualitative, exploratory, and future-oriented research design was applied, future-oriented in the sense that it examines how ongoing transformations are expected to evolve rather than only their current state. Interviews with 18 respondents were conducted between March and May 2026 with six actor categories across the forestry value chain: forestry contractors, large forest companies, machine and technology suppliers, research and academia, industry organizations, and broader forestry service companies. The material was analysed through a Gioia-inspired abductive thematic analysis, and the roadmap was developed by interpreting the resulting aggregate dimensions as areas of contractor exposure, translating them into cross-cutting content areas, and organizing these into a three-phase iterative structure informed by the strategic roadmapping literature.

Findings - Three aggregate dimensions emerged as coherent transformation pressures on Swedish forestry contractors: technological and digital transformation, value-chain and business model transformation, and capability and professionalization transformation. The empirical material surfaced an asymmetric value-chain dynamic in which contractors carry friction that larger value-chain actors do not always perceive. From these dimensions, a three-phase iterative roadmap (evaluate current position, formulate strategic direction, implementation and reflection) was developed to support contractors in moving from mainly reactive adaptation toward more deliberate strategic positioning.

Theoretical contributions - The study contributes in two ways. First, it provides an integrated understanding of how ongoing transformations together reshape the strategic situation of Swedish forestry contractors, moving beyond individual-transformation studies to an actor-level account of how multiple pressures interact. Second, it extends the strategic roadmapping tradition into the context of SME-scale, capital-intensive service providers operating under

structural value-chain dependence, adapting the framework through iterative design, an operational appendix, and an industry-association facilitator role.

Practical contributions - The roadmap provides a structured framework for contractor-level strategic adaptation. It is intended to be useful at the contractor level for firms working through their own strategic position, and at the broader sector level for Skogsentreprenörerna (the Swedish industry association for forestry contractors), large forest companies, and technology suppliers.

Limitations of study - The study is qualitative and contextually Swedish, future-oriented in examining expected developments but not predictive, and the roadmap has not been tested through actual implementation. Contractor variation means individual application will be specific to each firm.

Key words - Forestry contractors; strategic roadmap; industry transformation; value-chain asymmetry; proactive strategic positioning; Swedish forestry.

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1. Introduction

Swedish forestry is undergoing a period of significant transformation. Digital technologies, artificial intelligence and data-driven planning tools are reshaping how forest work is planned, executed and reported (Holmström, 2020; Visser & Obi, 2021). Stepwise automation and mechanization continue to advance, and fossil-free and electrified machinery is being explored (Visser & Obi, 2021; Nordfjell et al., 2010). Sustainability, biodiversity and certification demands are tightening through both formal regulation and broader pressure from customers, environmental organizations and society at large (Lindahl et al., 2017). These pressures do not arrive separately; they accumulate, interact and reshape the conditions under which forestry work is carried out (Onufrey & Bergek, 2021).

Forestry holds a substantial industrial and economic position in Sweden, supplying raw material to the country's pulp, paper, sawn-wood and bio-energy industries and contributing to wider ambitions around the bioeconomy and the climate transition (Näyhä & Pesonen, 2014). It is supported by a complex value chain in which forest owners, large forest companies, machine and technology suppliers, certification organizations, regulators and operational service firms each play distinct, mutually dependent roles (Häggström et al., 2013; Kronholm et al., 2021). Among these actors, forestry contractors are among the most central and most exposed: independent firms that perform contracted services such as harvesting, forwarding, silviculture, site preparation and clearing, and that carry out a substantial share of Sweden's operational forestry work (Häggström et al., 2013). They operate where industry-wide change becomes daily operational reality, which makes the question of how their role evolves under ongoing transformation central to the sector as a whole.

However, the role of forestry contractors is increasingly challenged by these multiple ongoing transformations. Digitalization reaches contractors not as an abstract opportunity but as a fragmented set of customer-specific systems, each with its own login, format and reporting flow, and as a steady increase in the administrative burden that follows from documentation and reporting expectations (Holmström, 2020). Artificial intelligence and data-driven planning tools place new competence demands on contractors and shift parts of the operational decision space outside the contractor firm (Holmström, 2020; Visser & Obi, 2021). Stepwise automation

and remote operation alter the economics of the machine fleet contractors are expected to invest in, raising questions about which technology generation to commit to and on what financing terms (Visser & Obi, 2021; Nordfjell et al., 2010). Electrification and fossil-free machinery introduce further investment uncertainty in a setting where contractors already carry capital-intensive assets on margins largely set elsewhere (Kronholm et al., 2021; Eriksson & Sääf, 2017). Sustainability and certification requirements arrive at the contractor level as documentation, monitoring and operational practice that must be delivered without compromising day-to-day productivity (Lindahl et al., 2017). Contractor work is also reshaped by the conditions of the relationship with large forest companies, where contract terms, planning quality and reporting systems are defined largely by the customer rather than negotiated with the contractor (Erlandsson, 2013; Kronholm et al., 2021). Operator shortages and rising competence demands operate as a structural constraint on capacity and as a further pressure on how contractor firms recruit, train and retain personnel (Häggström et al., 2013; Kronholm et al., 2021). These pressures are not new in isolation; what is changing is their accumulation and interaction. A new digital reporting requirement can simultaneously become an administrative burden, a competence demand and a shift in the customer relationship. Similarly, an operator shortage can be understood as a labour-market problem, a constraint on technology adoption and a limit on scaling. Contractor experience is therefore one of multi-dimensional change, where responses to any one pressure are entangled with responses to the others.

Two features of the contractor role make this multi-dimensional change especially demanding to navigate. First, the great majority of Swedish forestry contractors are small and medium-sized enterprises holding capital-intensive assets and operating on tight margins with limited managerial slack (Kronholm et al., 2021; Eriksson & Sääf, 2017); strategic work must coexist with day-to-day operational responsibility, leaving modest resources for sustained strategic analysis. Second, contractors operate under contractual conditions largely defined by larger value-chain actors: contract terms, planning quality, digital reporting systems and sustainability expectations are typically set by the customer and reach contractors as conditions to implement rather than parameters to negotiate. Contractor adaptation therefore develops under conditions the contractor cannot fully determine, which makes contractors structurally more exposed than the actors who define those pressures and their strategic options narrower and more interdependent. This combination of SME constraints and value-chain dependence points to several research gaps in how contractors interpret and respond strategically to transformation.

A first research gap concerns the understanding of how ongoing transformations within the forestry industry impact forestry contractors. Existing research on Swedish forestry transformation tends to address the sector at the level of the industry, the bioeconomy, large forest companies or specific technologies, and the contractor sector is typically treated as one stakeholder group among several rather than as the focal actor whose strategic options are being examined (Näyhä & Pesonen, 2014; Onufrey & Bergek, 2021; Holmström, 2020; Visser & Obi, 2021; Lindahl et al., 2017). Studies that do focus on Swedish forestry contractors have examined sector development (Häggström et al., 2013), business models and profitability (Kronholm et al., 2021) and sector-level economic performance (Eriksson & Sääf, 2017), each describing important conditions in isolation. What is missing is an integrated, forward-looking account of how the multiple transformations that contractors face simultaneously combine and interact at the contractor level. The first research gap is therefore the absence of an integrated understanding of how ongoing transformations within the forestry industry together impact Swedish forestry contractors.

A second research gap concerns the structured support contractors have available for navigating these transformations. Even where individual challenges are recognized, guidance for how a contractor might clarify its current position, prioritize across simultaneous pressures and adapt over time is largely absent. Strategic roadmapping has been developed precisely as a methodological tradition for this kind of work, supporting organizations in structuring their strategic adaptation processes under conditions of uncertainty (Phaal, Farrukh & Probert, 2004; Phaal, 2015). Most roadmapping research, however, has developed in the context of large industrial firms or whole-sector transitions rather than small and medium-sized capital-intensive service providers, and it does not directly address the contractor's specific situation: limited time and resources for strategic work, dependence on larger value-chain actors, and the need to keep strategic activity proportionate to operational reality. The second research gap is therefore the absence of a strategic roadmap adapted to this context, one that can support forestry contractors in structuring their strategic adaptation under ongoing industry transformation.

The purpose of this study is to develop a strategic roadmap to support Swedish forestry contractors in maintaining competitiveness under changing industry conditions. This purpose is approached by answering the following two research questions:

1. How will ongoing transformations within the forestry industry impact Swedish forestry contractors?

2. How can a strategic roadmap be designed to support Swedish forestry contractors in adapting to these transformations?

To answer these questions, the study applies a qualitative, exploratory and future-oriented research design based on 18 semi-structured interviews with respondents across the forestry value chain, including forestry contractors, large forest companies, machine and technology suppliers, research and academia, industry organizations and broader forestry service companies. The empirical material was analysed through an abductive thematic analysis, from which three aggregate dimensions were identified: technological and digital transformation, value-chain and business model transformation, and capability and professionalization transformation. These dimensions form the empirical basis for the strategic roadmap developed in the study.

Building on these findings, the study develops a three-phase iterative roadmap for contractor adaptation. The roadmap supports contractors in evaluating their current position, formulating a strategic direction, and implementing and revising strategic priorities over time. In doing so, it is intended to help contractors move from mainly reactive adaptation toward a more structured approach to long-term competitiveness. The roadmap can be used by individual contractor firms, but may also support Skogsentreprenörerna, the Swedish industry association for forestry contractors, in a facilitator role by helping member firms work through the roadmap in a more accessible and coordinated way. The thesis thereby contributes by linking ongoing forestry industry transformation to contractor-level implications and by offering a practical framework for structuring strategic adaptation over time.

2. Theoretical Background

2.1 Swedish forestry contractors and forestry value chain transformation

Forestry contractors are independent firms that provide contracted forestry services, primarily in practical forest operations such as harvesting, forwarding, silviculture, site preparation and clearing, but also in closely related planning and inventorying tasks that support field operations (Häggström et al., 2013; Kronholm et al., 2021). This working definition is broad enough to cover the range of activities represented by Skogsentreprenörerna's membership, while distinguishing forestry contractors from forest owners, large forest companies, machine and technology suppliers, certification organizations and general forestry consultants.

Häggström et al. (2013) document the development of the Swedish forestry-contracting sector and emphasize that contractors carry out a substantial share of the operational forestry work in Sweden, with the shift toward an outsourced contractor model accelerating from the late twentieth century onward. Contractors function as operational service providers rather than owners of the forest resource: they translate forest-management plans, customer requirements, environmental considerations and technological capabilities into the field work that produces the raw material for downstream industries. Kronholm et al. (2021) characterize contractors as business actors with distinctive business models and profitability dynamics, underlining that contractor performance depends not only on operational skill or machine productivity but also on the firm’s capacity to manage capital-intensive assets, structure contracts with larger customers and navigate the value-chain conditions under which it operates. Eriksson and Sääf (2017), in a Skogforsk grey-literature analysis, similarly frame the sector as facing recurring economic and structural challenges around utilization, cost coverage and margin pressure. Together, this work positions the Swedish forestry contractor as an SME that is operationally central to the value chain but whose strategic options are shaped by capital intensity, customer concentration and conditions set by other actors in the chain (Cox, Sanderson & Watson, 2001).

These conditions are increasingly being reshaped by ongoing transformation within the Swedish forestry industry. Mature, resource-based industries of the kind Swedish forestry represents rarely transform through a single disruptive event; they change instead through

cumulative, interacting pressures that unfold over time and interact with existing resource positions, competitive positions and investment structures (Onufrey & Bergek, 2021; cf. Christensen, 1997). Forestry transformation extends beyond technology adoption. Näyhä and Pesonen (2014) describe strategic change in the Nordic forest industry as a process that encompasses resources, skills, collaboration, organizational culture and managerial behaviour, in addition to technological development. Lindahl et al. (2017) characterize Swedish forestry as a model that combines high industrial productivity with intensifying expectations on biodiversity, climate, recreation and other ecosystem services, asked to deliver more of everything simultaneously (Lindahl et al., 2017).

Several specific transformation pressures recur across the recent literature. Digital transformation in the Swedish forestry value chain has been characterized as an integration challenge as much as a technology challenge; its potential is substantial, but its realization is constrained by fragmented data flows, unclear data ownership and uneven uptake across actors (Holmström, 2020). Automation and robotics advance along a near-term, stepwise trajectory rather than through full unmanned operation, with teleoperation, operator support systems and selectively mechanized operations as the more realistic near-term opportunities (Visser & Obi, 2021). Sustainability and certification expectations continue to tighten through both formal regulation and broader pressure from customers and society (Lindahl et al., 2017). For the focal actor of this thesis, these pressures are not experienced as abstract sector-level processes. They are experienced as changing requirements for technology use, customer coordination, contractual exposure, operational efficiency, sustainability compliance and competence development, filtered through the contractor's specific value-chain position.

2.2 Strategic roadmapping

Roadmapping originated as a technology-planning approach in industrial firms, where it was used to align technology development with product and market evolution over time (Kostoff & Schaller, 2001). Phaal, Farrukh and Probert (2004) developed the modern formulation that has since become the methodological reference point for the field. They describe roadmapping as a flexible planning framework that connects current resources, conditions and capabilities to future-oriented goals through structured pathways for action under uncertainty. The roadmap is, in their treatment, both a process and an artifact: a way of generating shared strategic

understanding within an organization, and a visual representation of the resulting plan that supports communication and revision.

Subsequent work has broadened this formulation considerably (Phaal, Farrukh & Probert, 2006; Phaal & Muller, 2009). Phaal (2015) describes roadmapping as a general strategy-and-innovation alignment approach rather than only a technology-planning tool. The underlying logic remains the same, namely structured movement from current position to future direction to action pathways, but the content being roadmapped expands. This expansion matters for the present thesis: the roadmap developed in Chapter 5 is not a technology roadmap but a strategic roadmap for proactive contractor positioning under multi-dimensional transformation pressure, and the broader formulation provides the methodological warrant for that move.

Two recent contributions inform the specific form of the roadmap developed here. Frishammar and Parida (2018) develop a transformation roadmap for incumbent firms undergoing circular business model change. Their work demonstrates how qualitative case material can be translated into a structured sequence of phases, key activities and outcomes, and how the resulting roadmap can be presented as both an empirical synthesis and a practical tool. Linde, Sjödin, Parida and Gebauer (2021) develop a phase-based framework for evaluating digital business model opportunities; their structure foregrounds the role of structured deliberation in avoiding commitment to ill-suited strategic moves and in surfacing explicit assumptions before action is taken. Both works inform the phase structure and iterative logic of the roadmap presented in this thesis, while the empirical substance is grounded in the interview material described in sections 3.2 and 3.3.

2.3 Towards a strategic roadmap for forestry contractors

The two preceding sections position the Swedish forestry contractor as an SME-scale, capital-intensive service provider whose strategic options are shaped by value-chain dependence and who must operate under cumulative and interacting transformation pressures, and they establish strategic roadmapping as a methodologically defensible way of translating qualitative transformation analysis into a structured strategic artifact. Bringing these two streams together raises a specific question: what kind of strategic roadmap is appropriate for forestry contractors given their context?

Three contextual features shape the answer. First, contractor firms have limited time and managerial slack for strategic work alongside day-to-day operations, a constraint characteristic of small firms in general (Penrose, 1959; Nooteboom, 1994) and of Swedish forestry contractors in particular (Kronholm et al., 2021); a roadmap aimed at this actor therefore cannot be resource-intensive or assume the sustained strategic capacity of larger firms. Second, contractors do not control most of the conditions under which they operate, since contracts, planning quality, digital systems and sustainability expectations are largely defined by larger value-chain actors, so a roadmap must work within the constraints of the contractor's value-chain position rather than assume they can be set aside. In Porter's (1985) terms, the contractor's scope for competitive positioning is constrained by the structure of the chain itself, not only by the firm's own resources. Third, much of contractor work is genuinely reactive: forestry operations involve substantial day-to-day reactivity to weather, machine breakdowns, customer-driven planning changes and site conditions, and this reactivity is part of the craft rather than a strategic deficit. The strategic question is therefore not whether contractors should be reactive or proactive in the absolute, but whether reactivity also dominates in areas where deliberate preparation and prioritization would be possible. Drawing on the distinction between deliberate and emergent strategy (Mintzberg & Waters, 1985), the roadmap is designed to create enough deliberate strategic clarity to support necessary operational reactivity, rather than to replace reactivity with predetermined plans.

These features have a further implication for how a contractor roadmap can realistically operate. If individual contractor firms have only limited capacity for sustained strategic work, the question of facilitation becomes central to the roadmap's practical applicability. The Swedish contractor industry association, Skogsentreprenörerna, offers a structural position from which the roadmap can be made accessible to its member firms, supporting them in working through the phases in a coordinated way that reduces the strategic burden on the individual firm. Chapter 5 develops the roadmap as a three-phase iterative framework that reflects these features, and Section 5.5 together with Appendix A elaborates in depth how Skogsentreprenörerna can act as a facilitator in supporting contractor firms' use of the roadmap.

3. Method

3.1 Research approach

This study adopts a qualitative, exploratory and future-oriented research design, where future-oriented means that the study examines how ongoing transformations are expected to evolve and what this implies for contractors, rather than only describing the current state. The design is applied to examine how ongoing transformations in the Swedish forestry industry affect the strategic positioning of forestry contractors. The qualitative approach is appropriate because the research interest lies in understanding interpretations, perceptions and actor-specific implications of broader industry change. The study is exploratory because the strategic perspective of Swedish forestry contractors is comparatively underexplored in the existing literature (Eisenhardt, 1989), and future-oriented in that it asks how ongoing transformations are expected to evolve and how contractors can adapt to them (Rohrbeck & Kum, 2018). The study is not a forecast or scenario exercise; it examines how respondents understand ongoing transformation pressures and how these can be translated into a strategic roadmap.

Beyond describing transformation impacts, the analysis translates them into a strategic framework, namely the three-phase iterative roadmap presented in Chapter 5. The logic underlying the analysis is therefore abductive rather than purely inductive or deductive (Dubois & Gadde, 2002). Themes emerged through iterative movement between the interviews, the theoretical background and the roadmap objective of the study. In practice, this meant that emerging empirical patterns prompted returns to the literature and vice versa: for example, the recurring contractor accounts of friction in customer relationships led back to the value-chain literature for conceptual framing, while the roadmap objective shaped which empirical patterns were pursued in follow-up questions and developed into themes. Section 3.3.1 describes the concrete analytical steps.

The research context is the Swedish forestry industry, with particular attention to the conditions under which forestry contractors operate. The unit of analysis is the strategic positioning of Swedish forestry contractors under changing industry conditions, rather than any individual contractor firm or the Swedish forestry industry as a whole (Yin, 2018). The study was conducted in collaboration with Skogsentreprenörerna, the Swedish industry organization for

forestry contractors. The collaboration provided practical access to respondents and contextual understanding of contractor work, but Skogsentreprenörerna is not treated as a case company; the analysis, interpretations and conclusions remain the responsibility of the author.

3.2 Data collection

3.2.1 Sampling and respondents

Respondents were selected through purposive sampling (Patton, 2015) with the aim of building a multi-perspective understanding of contractor conditions. Six broad actor categories were targeted: forestry contractors, large forest companies, machine and technology suppliers, research and academia, industry organizations and broader forestry service companies. Each category contributes a different vantage point on contractor work, customer relationships, technological change, regulatory and sustainability pressure and capability development.

In total, eighteen respondents (R1 to R18) participated in the study. Table 1 presents the distribution across actor categories, the date and duration of each interview, the format in which the material was documented, and the way the interview is used in the analysis. Respondents are referred to in the findings by actor category and respondent ID rather than by personal name or organization, both for clarity and to support anonymity where this was requested.

Table 1. Overview of respondents

ID	Actor category	Date (2026)	Duration	Data format	Use in analysis
R1	Technology and equipment supplier	13/03	00:30:24	Recorded and transcribed, 5 727 words	Direct quotations
R2	Machine manufacturer	16/03	00:27:37	Interview notes	Triangulation only
R3	Research institute	17/03	00:36:16	Recorded and transcribed, 5 430 words	Direct quotations
R4	Large forestry company	18/03	00:30:32	Interview notes	Triangulation only
R5	Large forestry company	18/03	00:40:15	Recorded and transcribed, 6 863 words	Direct quotations
R6	Large forestry	20/03	00:27:15	Recorded and transcribed,	Direct quotations

	company			4 282 words	
R7	Research and academia	23/03	00:36:05	Recorded and transcribed, 7 494 words	Direct quotations
R8	Forest industry organization	27/03	00:43:35	Recorded and transcribed, 5 829 words	Direct quotations
R9	Machine manufacturer	13/04	00:31:58	Recorded and transcribed, 5 402 words	Direct quotations
R10	Forestry contractor	21/04	00:19:17	Interview notes	Triangulation only
R11	Contractor industry organization (Part of group interview)	23/04	01:46:10	Recorded and transcribed, 16 133 words	Direct quotations
R12	Contractor industry organization (Part of group interview)	23/04	01:46:10	Recorded and transcribed, 16 133 words	Direct quotations
R13	Contractor industry organization (Part of group interview)	23/04	01:46:10	Recorded and transcribed, 16 133 words	Direct quotations
R14	Forestry contractor	24/04	00:41:05	Recorded and transcribed, 5 051 words	Direct quotations
R15	Forestry contractor	01/05	00:12:53	Interview notes	Triangulation only
R16	Forestry contractor	05/05	00:17:18	Interview notes	Triangulation only
R17	Forestry service company	05/05	00:41:02	Recorded and transcribed, 7 168 words	Direct quotations
R18	Technology and equipment supplier	12/05	00:34:34	Recorded and transcribed, 4 990 words	Direct quotations

3.2.2 Semi-structured interviews

Data was collected through semi-structured interviews (Kvale & Brinkmann, 2015) conducted between March and May 2026. The interview guide covered five broad themes: ongoing industry transformation pressures, technology and digitalization in forestry work, value-chain relationships and contracting, contractor operating conditions, and respondent expectations about how these areas will develop in the medium term. The semi-structured format provided consistency across interviews while allowing follow-up questions tailored to each respondent's role and area of expertise.

Most interviews were conducted via video call, with a smaller number conducted by phone, depending on respondent availability and preference. Interview lengths ranged from approximately twelve minutes to nearly two hours, with most lasting between twenty-five and forty-five minutes. The variation in length reflects practical rather than analytical factors: the longest interview was a joint conversation with three representatives from Skogsentreprenörerna, reported as R11, R12 and R13 sharing one transcript in Table 1, while the shortest interviews were phone conversations with individual contractors whose availability was limited. Interview length is therefore not evenly distributed across the six actor categories, and shorter, note-based interviews were used for triangulation rather than quotation, as described below; the implications of this imbalance are addressed in Section 6.5.

Most interviews were audio-recorded with respondent consent and subsequently transcribed in full. A minority were documented through structured notes, either at the respondent's request, because audio recording was not feasible, or because the conversation was conducted by phone without recording. This distinction carries through to how the material is treated analytically: direct quotations in the findings are taken only from transcribed sources (R1, R3, R5 to R9, R11 to R13, R14, R17, R18), while note-based interviews (R2, R4, R10, R15, R16) support the analysis through triangulation but are not quoted verbatim. Interviews were conducted in Swedish; quotations used in the findings have been translated into English by the author.

3.3 Data analysis

3.3.1 Abductive thematic analysis

The empirical material was analysed through a Gioia-inspired abductive thematic analysis (Braun & Clarke, 2006; Gioia, Corley & Hamilton, 2013; Miles, Huberman & Saldaña, 2014). The Gioia inspired structure supports systematic movement from concrete empirical observations to higher-order conceptual structures while preserving the traceability of each analytical move. Combined with an abductive orientation (Dubois & Gadde, 2002), the analysis developed in iterative dialogue with the theoretical background and the roadmap objective of the study.

The analysis proceeded in four steps. First, transcripts and structured notes were reviewed to identify recurring observations relevant to contractor conditions and industry transformation. These first-order observations were captured in respondent-near language, preserving the original framing and noting the respondent ID. For example, a statement from a forestry contractor (R14), "It is essentially an oligopoly market with a few very large companies. If they could have a shared portal, it would help us contractors who work for different customers," generated two first-order observations: "few large customers dominate the market" and "contractors carry the cost of fragmented customer systems." A second example, from a representative of the contractor industry organization (R13), "There is more and more administration falling on contractors. All these environmental requirements and so on, it keeps piling up," generated the first-order observation "administrative load is accumulating at the contractor level," which was subsequently clustered into the second-order theme "Regulatory and administrative burden."

Second, first-order observations were grouped into second-order themes by identifying patterns across multiple respondents. Codes such as "few large customers dominate the market," "contractors carry the cost of fragmented customer systems" and similar observations from R6, R8 and R17 about buyer concentration and contractual exposure were clustered into the second-order theme "Contractor-customer value-chain dynamics." Third, second-order themes were aggregated into three aggregate dimensions: technological and digital transformation, value-chain and business model transformation, and capability and professionalization transformation. A fourth dimension on regulatory and sustainability requirements was considered but ultimately integrated into the value-chain dimension, because the empirical material framed these requirements primarily as operating-environment conditions reaching contractors through customer relationships and administrative burden. Fourth, the structure was refined through iterative movement between the empirical material, theory and the roadmap objective. The resulting data structure is presented in Section 4.1.

3.3.2 Roadmap development

The strategic roadmap presented in Chapter 5 was developed from the empirical findings in three steps. The three aggregate dimensions were first interpreted as areas of contractor exposure to industry transformation. These areas were then translated into seven cross-cutting roadmap content areas (technology, competence, customer collaboration and contracts,

operational efficiency, digital and data readiness, service scope and compliance), each grounded in one or more of the aggregate dimensions. Finally, the content areas were organized into a three-phase iterative roadmap (Evaluate current position, Formulate strategic direction, Implementation and reflection) informed by the strategic roadmapping literature (Phaal, Farrukh & Probert, 2004; Frishammar & Parida, 2018; Linde et al., 2021).

3.4 Quality improvement measures

Trustworthiness is assessed against the four criteria of Lincoln and Guba (1985). Credibility was supported by including multiple actor perspectives across the forestry value chain; where claims appeared in multiple respondents from different actor categories, analytical confidence is correspondingly stronger, and where claims appeared in only one respondent, this is noted. Transferability is supported by describing the Swedish forestry contractor context and respondent categories in sufficient detail for readers to assess relevance to other contexts. Dependability was supported by using a consistent interview guide, organizing transcripts and notes systematically and applying the same coding logic across the material. Confirmability is supported by a transparent evidence trail from interview content through first-order observations and second-order themes to aggregate dimensions and the roadmap; direct quotations are taken only from transcribed material with respondent IDs preserved throughout. While the coding and interpretation were undertaken by a single researcher and therefore did not involve inter-coder agreement, steps were taken to strengthen the credibility and transparency of the analysis. These included maintaining a clear evidence trail from participant quotations to first-order observations and aggregate dimensions, and discussing the emerging data structure with the supervisor.

3.5 Ethical considerations and AI usage

Respondents were informed of the purpose of the study and the intended use of the interview material before each interview, and participation was voluntary throughout. Audio recording was conducted only with explicit consent. Respondents are presented using respondent IDs and broad actor categories rather than personal names, and details that could identify a respondent without contributing to the analytical argument have been omitted. Interview material has been stored and handled solely for the purpose of this thesis and is treated as confidential.

Claude Sonnet 4.6 was used in a structured way to support structural planning, language refinement, alternative formulations during drafting, and internal coherence checking across chapters. AI was not used to conduct the empirical analysis independently. Coding of the interview material, identification of themes and the interpretive moves linking themes to aggregate dimensions and the roadmap were performed by the author. Empirical claims have been verified against the underlying interview material, and references have been checked against the cited sources. The author retains full responsibility for the content of the thesis.

4. Findings

This chapter presents the findings of the study. Sections 4.2 through 4.4 answer RQ1 by showing how ongoing transformations in the Swedish forestry industry impact forestry contractors across three aggregate dimensions identified in the analysis. Chapter 5 then answers RQ2 by translating these findings into the strategic roadmap that constitutes the principal output of the thesis. All interview material was originally in Swedish. Direct quotations have been translated into English and lightly edited for readability by removing filler words, repetitions, and transcription artifacts, while preserving the substantive meaning of the respondent's statement. Direct quotations are drawn only from recorded and transcribed interviews; the notes-based interviews (R2, R4, R10, R15, R16) support the analysis through triangulation but are not quoted verbatim.

4.1 Overview of empirical themes

The abductive thematic analysis described in Chapter 3 produced a structured account of how ongoing transformations in Swedish forestry are reaching the contractor level. Across eighteen interviews with contractors, large forest companies, machine and technology suppliers, research and academia, and industry organizations, three aggregate dimensions emerged as coherent transformation pressures on Swedish forestry contractors: technological and digital transformation, value-chain and business model transformation, and capability and professionalization transformation.

The dimensions are analytically distinct but interrelated in the contractor's operating reality. Customer-driven digital systems are simultaneously a technological pressure and an administrative burden. The operator shortage is at once a labour-market phenomenon, a competence challenge, and a constraint on technology adoption. Sustainability and certification requirements reach contractors not as abstract policy but through customer contracts, documentation routines, and operational conditions. The separation into three dimensions is made for clarity of presentation, not because the empirical material sits cleanly in non-overlapping bins. Where intersections shape the contractor experience, they are made explicit in the relevant section.

Figures 1, 2, and 3 present the resulting data structure, showing the progression from first-order observations through second-order themes to aggregate dimensions. The figures are organized according to the Gioia-inspired analytical approach described in Section 3.3.1 and provide the evidence trail for the analytical claims made in the remainder of this chapter.

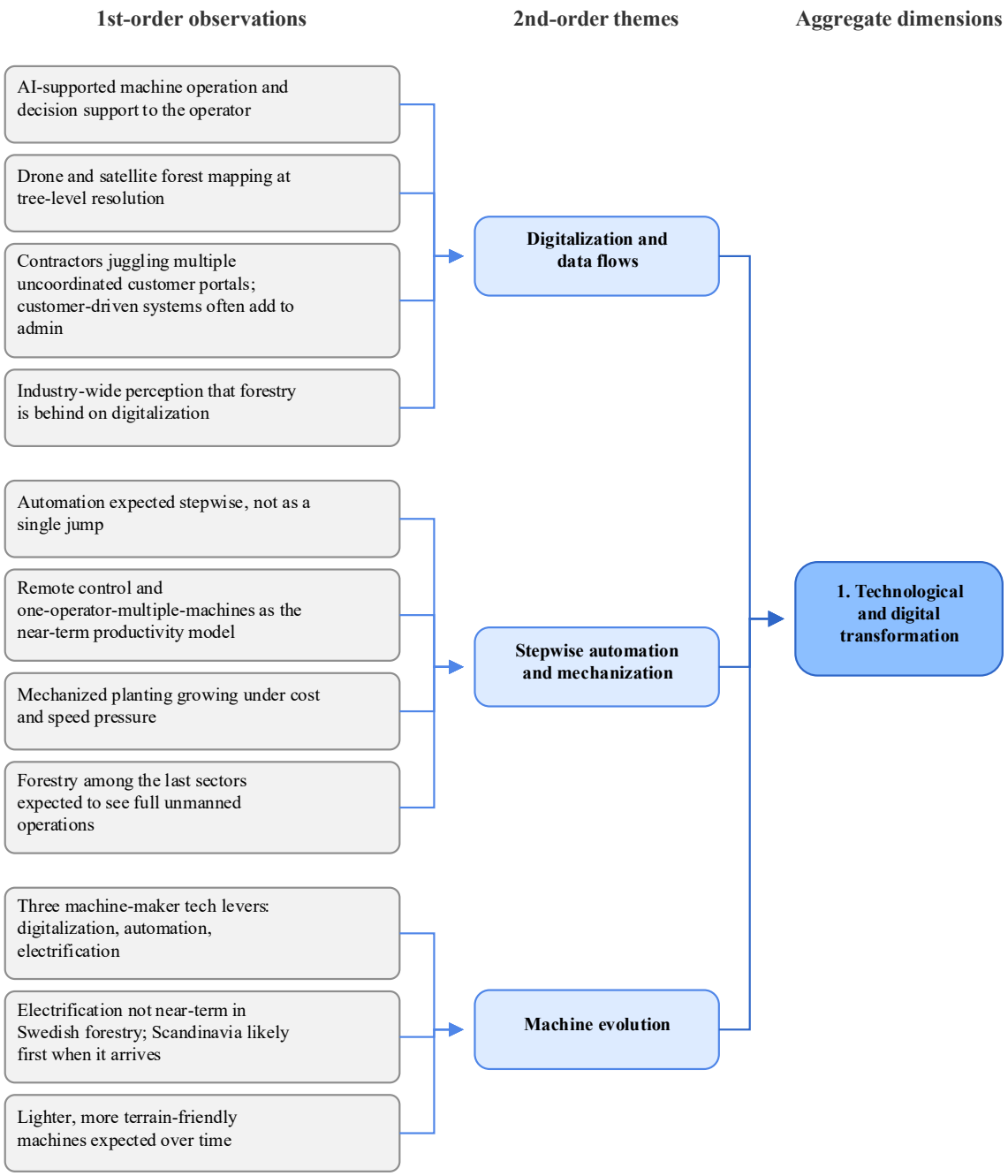


Figure 1. Data structure: Technological and digital transformation

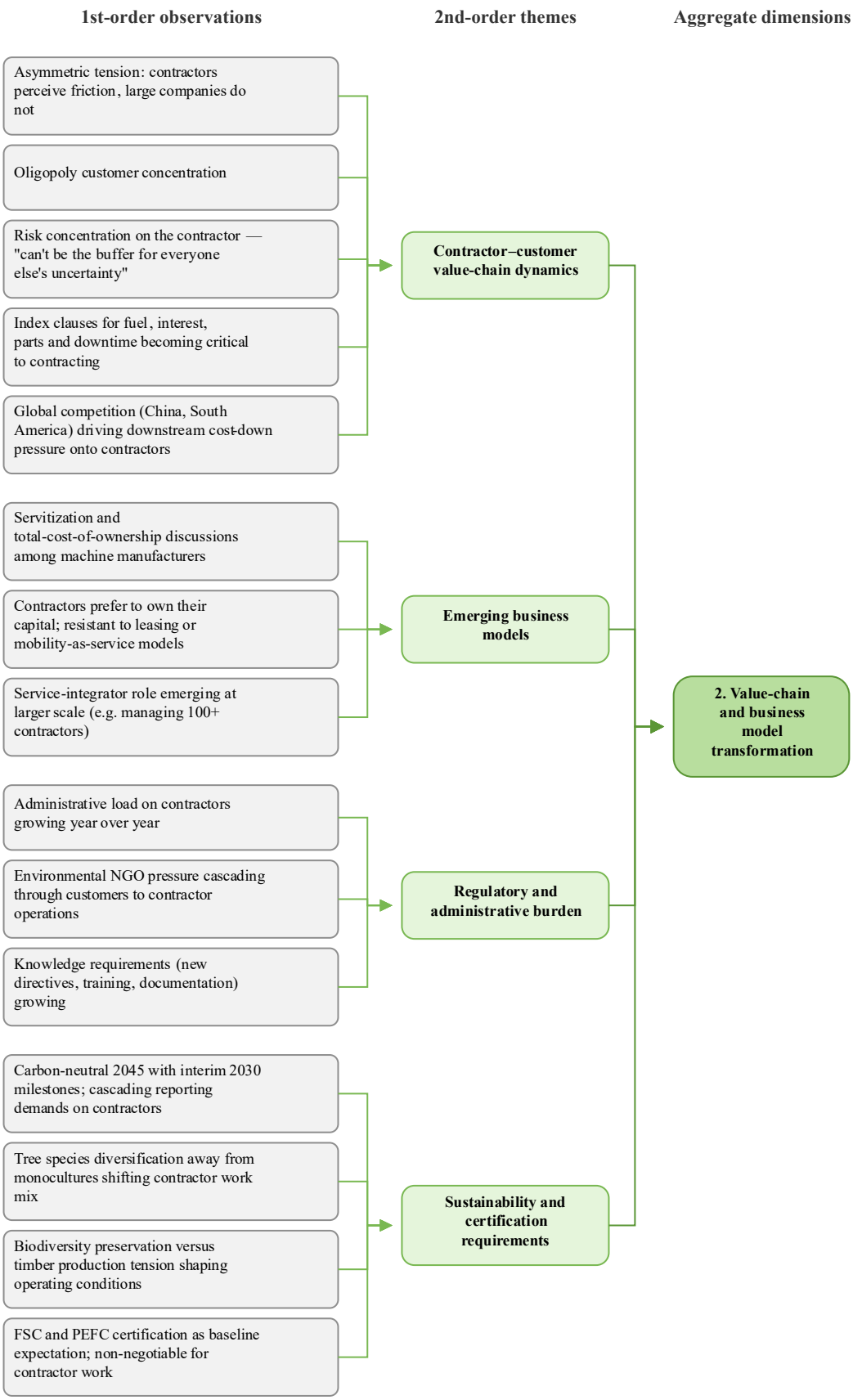


Figure 2. Data structure: Value-chain and business model transformation

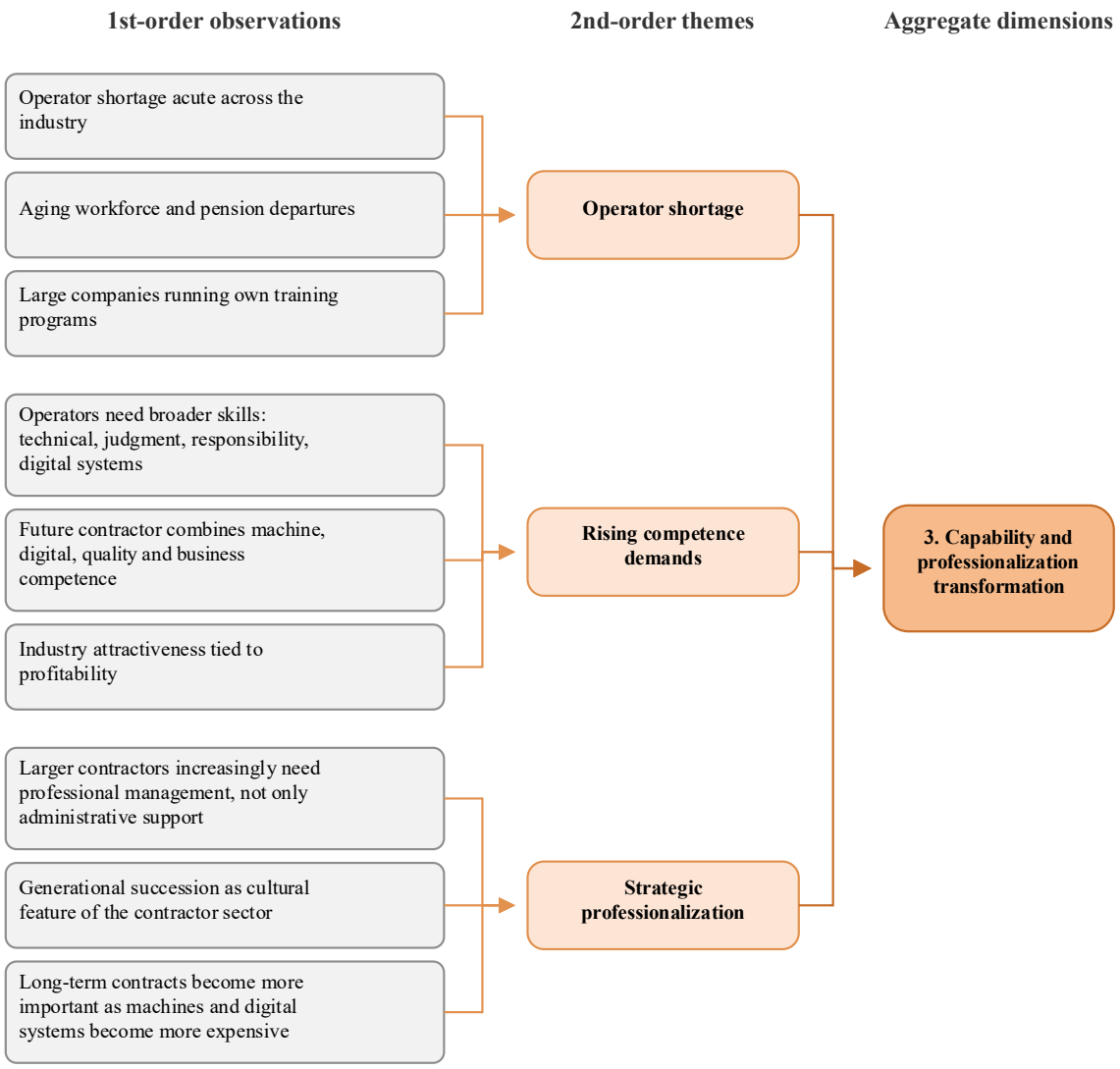


Figure 3. Data structure: Capability and professionalization transformation

Sections 4.2 to 4.4 present each aggregate dimension in turn, organized around its constituent second-order themes. Each subsection states the central observation, grounds it in representative observations from the transcribed interviews, and closes with a brief synthesis of what the theme implies for the contractor's position. Chapter 5 then synthesizes the three dimensions into the strategic roadmap that constitutes the central output of the thesis.

4.2 Technological and digital transformation

Technological and digital transformation reaches Swedish forestry contractors along three interlocking fronts: through the digitalization of forestry information flows, through the stepwise progression of automation and mechanization in field operations, and through broader machine evolution. Technological change in forestry is real, but its contractor-level impact depends on the conditions under which adoption takes place, including who controls the data, how systems integrate across customers, how the economics of new machines are structured, and whether technology delivers value back to the contractor.

4.2.1 Digitalization and data flows

Digitalization in Swedish forestry is already substantial. Respondents from across the value chain describe a sector in which planning, harvesting, transport, and downstream processing are increasingly integrated through digital systems and data flows. A representative from a technology and equipment supplier (R1), describing how digitalization now spans the full forestry value chain, observed that the change is no longer a future prospect but an operating reality:

> *"Every chain in the entire forestry industry, from the planning of an object at the very beginning, it is followed all the way until the timber arrives at the sawmill yard."* (R1, technology and equipment supplier)

The integration that this statement describes is, however, experienced unevenly at the contractor level. Two patterns recurred across the contractor and industry-organization respondents. The first concerns the proliferation of customer-specific digital systems, each requiring its own login, format, and reporting flow. A representative from a forestry contractor (R14) described the practical effect of this fragmentation directly:

> *"We contractors live in many systems. The big forest companies have their own customer portals, and we have to go into the customer's portal and report into their system."* (R14, forestry contractor)

For the contractor, digitalization in this pattern arrives as administrative overhead rather than operational benefit. Each customer relationship adds another system that the contractor's organization must learn to work within, and the integration across those systems sits with the contractor rather than with the customers that designed them.

The second pattern concerns ownership and use of the data that contractor machines generate during their work. A representative from the contractor industry organization (R11) treated this as an unresolved structural question that runs through much of the broader digitalization debate:

> *"The question of machine data is very central to all of this, and one of our challenges. Who owns the data?"* (R11, contractor industry organization)

Taken together, these observations identify a digitalization paradox at the contractor level. More data is being generated, and more digital systems exist than ever before, but the integration that would translate this into operational benefit for the contractor remains incomplete, and the questions of data control and burden have not been settled. Digitalization is not, in the empirical material, a unitary phenomenon that affects contractors uniformly. It is a set of conditions whose impact depends on how access, integration, and ownership are configured.

4.2.2 Stepwise automation and mechanization

Where digitalization is already operationally present, automation and mechanization are described as developing more gradually. A consistent view across machine manufacturers, research and academia, and industry organizations is that the most plausible near-term trajectory for forestry automation is stepwise rather than disruptive. A respondent from research and academia (R7), discussing what limits productivity gains in current operations, framed the logic in economic terms:

> *"I believe it will go that way step by step. What creates an economic problem today is that we have a one-to-one relationship between the human and the machine."* (R7, research and academia)

The direction of development that follows from this constraint is specific. Rather than a leap to fully unmanned operation, the near-term automation gain comes from breaking the one-to-one ratio between operator and machine. The same respondent continued by identifying remote operation as the most realistic first step:

> *"If we could get one person to drive two machines, we would have completely different conditions. To begin with, I believe in remote operation."* (R7, research and academia)

The view from the forest-industry organization (R8) reinforced the stepwise picture by adding an important boundary. Even as automation advances, forestry is unlikely to reach a state where all operational judgement can be removed from the field:

> *"I do not believe in fully self-driving vehicles. There will still need to be people on site to be able to solve certain problems."* (R8, forest industry organization)

For the contractor, these accounts describe automation not as an imminent disruption but as a gradual reshaping of where machine cost and human skill are concentrated. The most concrete near-term mechanization development described by respondents is in planting work, where mechanized solutions are emerging under cost and labour-shortage pressure. The strategic question for contractors is therefore not whether automation arrives, but how to invest, train, and contract under conditions of stepwise change rather than abrupt disruption.

4.2.3 Machine evolution

Beyond automation, respondents described broader machine evolution as a transformation pressure with direct economic consequences for contractors. The historical trajectory of forestry machinery has been one of incremental productivity gains. Several respondents suggested that the rate of leap-style improvement is slowing, and that the next phase of change will be felt as much in machine economics as in machine capability. A representative from a technology and equipment supplier (R1), describing the limits of the historical leap pattern, framed the shift directly:

> *"There is a belief that the technological leaps we have made historically in forestry will continue. I think people are starting to realize now that we cannot make such large leaps anymore."* (R1, technology and equipment supplier)

At the same time, the machines that contractors are expected to invest in are becoming more sophisticated and more expensive. A representative from a machine manufacturer (R9) connected this trajectory directly to the contractor's business-model position:

> *"Machines with more autonomy and perhaps electrification will become significantly more expensive. The business model may become more OPEX-based rather than CAPEX-based in the future."* (R9, machine manufacturer)

Adoption is also conditional on the surrounding ecosystem. For mechanized planting in particular, a representative from a technology and equipment supplier (R18) noted that adoption depends on conditions outside the contractor's direct control:

> *"There is much that has to be standardized and stabilized and trained. But if there is drive from the larger players, then it will happen fast."* (R18, technology and equipment supplier)

Machine evolution thus affects contractors through a combination of investment exposure, business-model pressure, and adoption-condition uncertainty. The contractor is not simply choosing whether to adopt new machinery. The conditions under which adoption is economically and operationally viable are being shaped by larger value-chain actors and supplier strategies, which means contractor strategy on machinery must respond not only to the technology itself but to the ecosystem within which it becomes usable.

4.3 Value-chain and business model transformation

If technological transformation describes how the operational and informational environment around contractors is changing, value-chain and business model transformation describes how the conditions under which contractor work is bought, coordinated, documented and valued are changing. This dimension emerged as the analytical centre of the empirical material and most clearly surfaces the asymmetric structure of contractor experience: the conditions under which contractors operate are shaped largely by actors elsewhere in the chain, and the friction contractors experience is not always perceived in the same way by the actors generating it. It is presented across four themes rather than three, reflecting both its larger share of the empirical material and the decision in Section 3.3.1 to integrate regulatory and sustainability requirements here: contractor-customer value-chain dynamics, emerging business models, regulatory and administrative burden, and sustainability and certification requirements.

4.3.1 Contractor-customer value-chain dynamics

The relationship between forestry contractors and their primary customers, the large Swedish forest companies, was discussed by virtually every respondent. From the contractor side, the relationship is structured by buyer concentration, capital intensity, and dependence on the operating conditions that customers set. A representative from a forestry contractor (R14), describing the structural shape of the market, named the dynamic plainly:

> *"It is essentially an oligopoly market with a few very large companies. If they could have a shared portal, it would help us contractors who work for different customers."* (R14, forestry contractor)

The economic exposure that follows from this structure was articulated forcefully by a representative from the contractor industry organization (R13). The combination of high capital commitment and dependence on continuous utilization translates contract conditions directly into firm survival:

> *"You have invested ten or twelve million in a harvesting team. They need to run eleven months a year. If they stop for a month, it is essentially bankruptcy for many."* (R13, contractor industry organization)

Respondents from the research community framed the same dynamic in terms of contract design and the conditions surrounding the contracted work. As a respondent from a research institute (R3) put it, price alone is an insufficient description of what contractors are being paid for:

> *"The agreed price is one thing, but the conditions given for the work are at least as important."* (R3, research institute)

It is important to note that this dynamic is not experienced symmetrically. The large forest companies represented in the study described the contractor relationship in partnership terms and emphasized shared goals. A respondent from a large forestry company (R5) framed the desired relationship as a collaborative one:

> *"A contractor's role is really that we want them as a collaboration partner. We want to be able to establish an agreement where we have an understanding."* (R5, large forestry company)

The contractor side and the customer side thus describe the same value chain in markedly different terms: contractors and the contractor industry organization emphasize buyer concentration, capital exposure and the conditions surrounding the contracted work, while the large forest companies describe the relationship in partnership terms and emphasize shared goals. Neither account describes the other side as acting in bad faith; the friction that contractors report in planning and contract conditions did not appear with the same weight in the customer-side accounts. The interpretation of this asymmetric pattern, and its consequences for how the roadmap is framed, is developed in Section 6.2.

4.3.2 Emerging business models

Alongside changing contract conditions, several respondents described emerging shifts in how contractor business models may be configured in the medium term. The shift discussed most consistently is from capital ownership toward usage-based or outcome-based models. A representative from a machine manufacturer (R9), looking forward to where the underlying business model of forestry equipment may move, described one variant of this shift:

> *"The contractor might, instead of buying and owning a machine, pay for use, or pay for a certain outcome."* (R9, machine manufacturer)

The same respondent immediately introduced a counter-pressure that limits how widely such models can spread. Smaller contractors in particular value ownership and direct financial control of their assets, which works against the spread of servitization-style models:

> *"Particularly the smaller contractors are quite inclined to have full control over their business and prefer self-financing."* (R9, machine manufacturer)

A different form of business-model change emerged from the supplier side. A representative from a technology and equipment supplier (R18), describing the conditions under which mechanized planting contractors operate profitably, suggested that the most successful contractors are increasingly those who broaden their service from a discrete operational task to a more holistic one:

> *"The most profitable contractor has the best understanding of the whole, not just pushing a seedling into the ground, but understanding that they are selling new forest."* (R18, technology and equipment supplier)

Across these accounts, emerging business models read best not as a universal prescription for all contractors but as a set of possibilities and pressures. Service-based models, outcome-based contracts, and broader service scope are present in supplier and customer discourse, but the contractor's structural position (capital intensity, preference for ownership, customer dependence) shapes which options are economically feasible and strategically attractive in any given firm.

4.3.3 Regulatory and administrative burden

A third theme within value-chain transformation concerns the growing regulatory and administrative load that reaches contractors through customer relationships, certification

systems, and broader industry expectations. A representative from the contractor industry organization (R13) characterized the trajectory clearly when describing how new requirements accumulate at the firm level:

> *"There is more and more administration falling on contractors. All these environmental requirements and so on, it keeps piling up."* (R13, contractor industry organization)

Within this growing administrative load, two specific patterns emerged. The first is the proliferation of customer-specific reporting systems, mirroring the digitalization fragmentation described in Section 4.2.1 but viewed through the lens of administrative cost. A representative from a forestry contractor (R14) described the practical consequence:

> *"We have our own system, but then we have to go into the customer's portal and report into their system."* (R14, forestry contractor)

The administrative burden is not contested on the customer side, though it is framed differently. A respondent from a large forestry company (R5) acknowledged the issue directly from the buyer's perspective:

> *"We sit with a great deal of administration. Some of our contractors are not particularly enthusiastic about administering things."* (R5, large forestry company)

The regulatory and administrative dimension thus reaches contractors as concrete operational workload rather than as abstract requirements. Documentation, certification paperwork, reporting cycles, training requirements, and compliance routines accumulate at the firm level, often without corresponding compensation or administrative simplification. This makes administrative burden not merely an inconvenience but a real factor in contractor margin pressure and capacity allocation.

4.3.4 Sustainability and certification requirements

Sustainability and certification requirements form a closely related but conceptually distinct second-order theme within the value-chain dimension. Where the administrative burden of Section 4.3.3 is the operational workload that documentation creates, sustainability requirements describe the substantive content of what contractors are being asked to deliver,

monitor, and comply with. A representative from a large forestry company (R6), describing how customer-side sustainability demands have evolved, summarized the direction concisely:

> *"Sustainability requirements are very tangible now and have gradually been tightened more and more."* (R6, large forestry company)

The same respondent also articulated an important distinction. Formal compliance with national legislation and certification systems is no longer sufficient to settle the sustainability question, because broader societal and stakeholder expectations apply pressure beyond formal frameworks:

> *"It does not matter that we follow Swedish legislation to one hundred percent and that we are certified. We still face quite hard pressure from environmental organizations."* (R6, large forestry company)

From the broader-service perspective, sustainability is visible as a concrete change in what work contractors are being asked to perform. A respondent from a larger forestry service company (R17) connected the sustainability shift to changes in tree-species composition and forest resilience:

> *"We have to create more resilient forests, which means a greater diversity of tree species in the forests."* (R17, forestry service company)

Sustainability and certification thus reach contractors in three reinforcing ways: as documentation and compliance load, as substantive operational requirements on how forestry work is done, and as exposure to societal and stakeholder pressure that runs through their customer relationships. For the contractor, these requirements function less as a discrete external constraint than as an integral part of the operating environment within which competitive positioning has to be developed.

4.4 Capability and professionalization transformation

The third aggregate dimension concerns the people, competences, and organizational structures that contractors need to remain competitive under the conditions described in Sections 4.2 and 4.3. Contractor capacity is increasingly shaped by capability and professionalization questions rather than by machine productivity alone. The dimension is presented across three themes: operator shortage, rising competence demands, and strategic professionalization.

4.4.1 Operator shortage

Operator shortage was raised by virtually every contractor respondent and corroborated from the customer and industry-organization side. It was described not as a transient labour-market problem but as a structural capacity constraint on the sector. A representative from the contractor industry organization (R13), framing the shortage in quantitative terms, captured the underlying mismatch between training capacity and demand:

> *"Around eight hundred new machine operators are needed each year, and roughly three hundred and fifty are trained, of which not all end up in the forestry sector."* (R13, contractor industry organization)

The shortage is not contested on the customer side. A respondent from a large forestry company (R5) characterized competence supply as the most serious medium-term challenge they see:

> *"If there is something that hangs like a dark cloud over us, it is competence supply, particularly for machine operators."* (R5, large forestry company)

The shortage is not solely a function of training capacity. A respondent from a technology and equipment supplier (R1) located the recruitment difficulty also in the comparative attractiveness of forestry as an employment sector:

> *"There is quite a difference between the IT sector and the forestry sector when it comes to finding the people you want. In the forestry sector it is difficult."* (R1, technology and equipment supplier)

Operator shortage thus functions as a structural ceiling on contractor capacity. It also intersects with the other dimensions: technology adoption that requires more capable operators becomes harder when operators are scarce, and customer demands for higher-quality field execution become harder to meet when the recruitment base is shrinking.

4.4.2 Rising competence demands

Beyond the quantitative shortage of operators, respondents described a qualitative change in what an operator role actually entails. A representative from the contractor industry organization (R13), emphasizing how broad the modern operator role has become, described the scope of competence required today:

> *"A machine operator today, it is an incredible education. Environmental requirements, nature considerations, you cannot just take anyone and put them in."* (R13, contractor industry organization)

The competence shift includes a digital dimension. A respondent from a machine manufacturer (R9) connected operator competence directly to the digital tools that increasingly mediate the operator's work:

> *"The important first step is to engage with the digital aids and digital tools. You have to be able to work with the systems."* (R9, machine manufacturer)

Looking further ahead, a respondent from research and academia (R7) suggested that the skill profile rewarded in operator roles may itself shift as technology develops:

> *"From having been the most skilled driver, perhaps it is other abilities that will be valued in the future."* (R7, research and academia)

Competence demands thus broaden along multiple axes simultaneously: technical machine skill, environmental and quality judgement, digital systems literacy, administrative competence, and communication. For the contractor, rising competence demands have two implications. They make the existing operator shortage harder to resolve through volume alone, because the kind of person being recruited has to bring a wider skill set, and they make the contractor firm itself more dependent on developing competence internally rather than relying on a fully-formed labour supply from outside.

4.4.3 Strategic professionalization

The third theme within this dimension concerns the contractor firm itself rather than its individual operators. Several respondents described an evolution in what it takes to operate a contractor firm well, moving from a practical-craft focus on doing forestry work to a more managerial focus on running a business that does forestry work. A representative from a technology and equipment supplier (R1), articulating the direction of this shift, captured it succinctly:

> *"The direction is increasingly that forestry contractors become business owners, rather than that they remain machine operators."* (R1, technology and equipment supplier)

This professionalization is not a critique of how contractors currently operate. The empirical material was clear that contractor expertise in practical forestry remains the foundation of the sector. Rather, the shift is one of adding business-management competence on top of the operational core. A representative from the contractor industry organization (R12) framed the shift as a synthesis:

> *"It is not just about doing the work but also about being a business owner. Otherwise, you can work as much as you want and still not succeed."* (R12, contractor industry organization)

The professionalization shift also reframes the contractor's relationship with customers. A representative from a large forestry company (R5) connected professionalization directly to the partnership tone described in Section 4.3.1:

> *"We have to become collaboration partners. We have to become equals and each be able to carry our part of the agreement together."* (R5, large forestry company)

Strategic professionalization is therefore not about contractors becoming something they currently are not. It is about contractors adding deliberate strategic and business-management work to the operational expertise that already defines the sector. This shift is the foundation on which the proactive contractor positioning developed in the roadmap in Chapter 5 becomes possible.

5. A Strategic Roadmap for Forestry Contractor Adaptation

Sections 4.2 through 4.4 established that Swedish forestry contractors face transformation pressures along three reinforcing dimensions. The interview material also made clear that these pressures are typically experienced reactively rather than navigated proactively. This chapter synthesizes the findings into a strategic roadmap that supports contractors in moving from reactive adaptation toward more deliberate strategic positioning. The roadmap is the principal output of the thesis and directly answers RQ2.

5.1 Roadmap overview and development

The format draws on the strategic roadmapping tradition discussed in Section 2.2 (Phaal, Farrukh & Probert, 2004; Phaal, 2015) and on phase-based transformation frameworks in adjacent contexts (Frishammar & Parida, 2018; Linde et al., 2021). The substance is derived from the interview material and shaped by the asymmetric value-chain conditions identified in Section 4.3.1.

The three aggregate dimensions identified in Chapter 4 enter the roadmap in two explicit ways. First, they define the content of the diagnosis: in Phase 1 the contractor maps its exposure to technological and digital, value-chain and business model, and capability and professionalization transformation, and the reflection questions in Appendix A are organized by these same three dimensions. Second, the dimensions were translated into the seven cross-cutting content areas that structure prioritization in Phase 2, following the development logic described in Section 3.3.2. The aggregate dimensions are therefore not background to the roadmap but its analytical backbone: they determine what is diagnosed in Phase 1, prioritized in Phase 2 and reflected against in Phase 3. Figure 4 presents the roadmap visually, and Table 2 elaborates each phase along three dimensions: its purpose, its key activities and its expected deliverables.

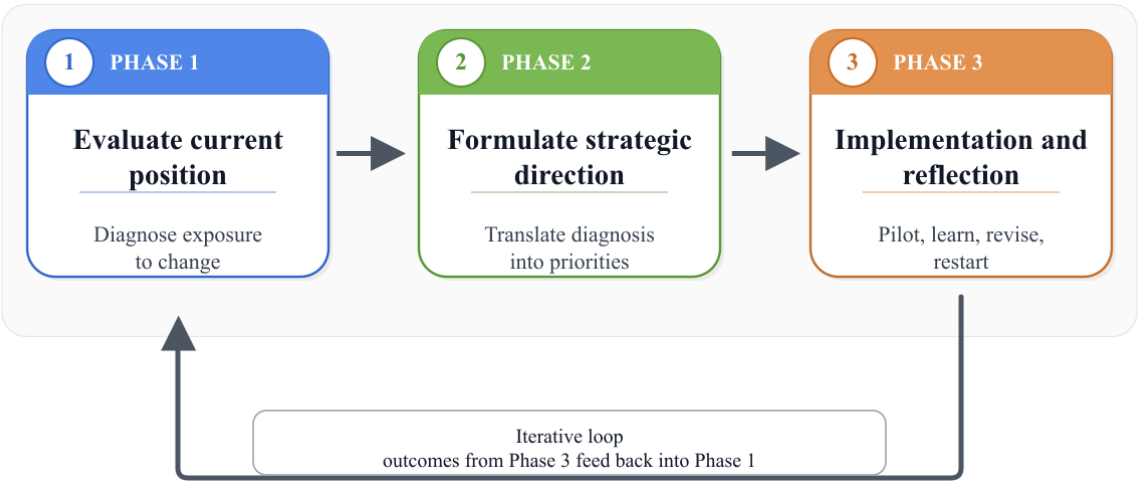


Figure 4. Three-phase strategic roadmap for forestry contractor adaptation

Table 2. Phase elaboration

	Phase 1 Evaluate current position	Phase 2 Formulate strategic direction	Phase 3 Implementation and reflection
Purpose	Create a structured diagnosis of the contractor's current position and exposure to industry transformation.	Translate the diagnosis into prioritized strategic actions for the contractor.	Move from plan to action while continuously learning and revising the roadmap.
Key activities	• Map ongoing transformations (i.e. the three aggregate dimensions) and their effects on contractor operations • Assess current capabilities, contract terms, and digital readiness • Identify strategic gaps and emerging opportunities	• Prioritize across technology, competence, customer collaboration, and operational areas • Decide service-scope, partnership, and contract-design choices • Define rationale and expected outcomes for each priority	• Pilot selected actions and track outcomes • Reflect internally with employees and externally with customers and partners • Revise priorities and re-enter Phase 1
Deliverables	Current-position diagnosis covering operational, contractual, capability and digital aspects.	Strategic action plan with prioritized actions, rationale, and expected outcomes per action area.	Implemented actions, documented outcomes, revised priorities, and an updated roadmap cycle.

5.2 Phase 1: Evaluate current position

Phase 1 produces a structured diagnosis of the contractor's current position and exposure to ongoing industry transformation. Across the interview material, contractors typically respond to pressures as they arrive rather than from a prior structured view of where the firm stands. Diagnosis is the prerequisite for any shift toward proactive behaviour.

The phase comprises three key activities. The first activity is to map ongoing transformations, that is, the three aggregate dimensions identified in Chapter 4, and their effects on the firm's operations. This mapping is carried out because the pressures reach each contractor differently depending on its customers, service mix and region, and a firm cannot prioritize against pressures it has not made explicit. In practice, it can be conducted as a structured review session within the firm, working through the reflection questions in Appendix A dimension by dimension. The second activity is to assess the firm's current capabilities, contract terms and digital readiness against those pressures. This assessment is carried out because exposure is a function of the firm's own position, not of the pressure alone; the same customer-system change can be trivial for one contractor and critical for another. In practice, it involves concrete steps such as listing the customer systems the firm operates within and the administrative time they consume, noting when each major contract comes up for renewal, and reviewing expected operator turnover. The third activity is to identify strategic gaps, where the firm's position is misaligned with where pressures are heading, and emerging opportunities, where the firm's position offers an undeveloped advantage. This step is carried out because the diagnosis must end in judgements that can be prioritized, not only in description. In practice, it is supported by completing the four-column diagnosis template in Appendix A for each aggregate dimension in turn.

The expected outcome of Phase 1 is a current-position diagnosis covering the operational, contractual, capability and digital aspects of the firm, structured by the three aggregate dimensions. The diagnosis distinguishes pressures that need attention in the next twelve to eighteen months from those that can be revisited in a later cycle, and it constitutes the direct input to the prioritization work of Phase 2.

5.3 Phase 2: Formulate strategic direction

The purpose of Phase 2 is to translate the Phase 1 diagnosis into a small set of prioritized strategic actions for the firm. The empirical material makes clear that without explicit prioritization the default is reactive response to whichever pressure is loudest in any given week, which is precisely the pattern the roadmap is designed to counteract.

The phase comprises three key activities. The first activity is to prioritize across the seven cross-cutting content areas that surfaced from the analysis: technology, competence, customer collaboration and contracts, operational efficiency, digital and data readiness, service scope and compliance. Prioritization is carried out because not all areas can receive deliberate strategic attention in a single cycle, and choosing which will, and which explicitly will not, is what separates a strategic direction from a wish list. In practice, the firm selects two or three content areas for the coming cycle and records which areas are consciously deprioritized for later cycles, supported by the Phase 2 reflection questions in Appendix A. The second activity is to make explicit choices on service-scope, partnership and contract-design questions. These choices are treated separately because they tend to require longer commitment than other actions and are difficult to reverse once made. In practice, this can mean deciding before a major contract renewal whether to broaden the service offer toward planning or silviculture, whether to pursue closer collaboration with a specific customer, or which contract terms to negotiate differently. The third activity is to define the rationale and expected outcome for each prioritized action, making implicit assumptions visible so that Phase 3 has something concrete to evaluate against. This step is carried out because actions whose success criteria were never articulated cannot generate learning, and it draws on the structured-deliberation logic in Linde et al. (2021). In practice, the firm completes the strategic action plan template in Appendix A, recording action, rationale, expected outcome and time horizon for each priority.

The expected outcome of Phase 2 is a strategic action plan with prioritized actions, the rationale behind each action and the outcome expected if the action succeeds, organized per content area. The plan also records the assumptions underlying the prioritization, which gives Phase 3 a concrete basis for evaluation and revision.

5.4 Phase 3: Implementation and reflection

The purpose of Phase 3 is to move from plan to action while building continuous learning into the cycle. The roadmap is iterative rather than linear because forestry transformation is ongoing and uneven; Phase 3 therefore does not produce a final position but a new and better-informed starting point for Phase 1.

The phase comprises three key activities. The first activity is to pilot selected actions and track their outcomes. Piloting is carried out because implementing on a limited scope keeps risk proportionate to the resources of an SME-scale firm while still generating real evidence. In practice, the firm implements an action on a defined scope, for example one customer relationship, one machine team or one season, and records both intended and unintended outcomes in the tracking template in Appendix A. The second activity is to reflect on those outcomes internally with employees and externally with customers and partners. Reflection is carried out because outcomes only become learning when they are interpreted, and because how an action lands with customers and operators reveals the value-chain conditions the firm operates within. In practice, this can take the form of a short internal review session with the employees involved and a follow-up dialogue with the customer or partner affected by the action. The third activity is to revise priorities and re-enter Phase 1 with an updated sense of position and exposure. Revision is carried out because conditions shift between cycles and a plan that is never revised reverts to being a document rather than a process. In practice, the firm decides for each action whether to carry it forward unchanged, revise it or retire it, and notes whether its overall exposure has shifted in ways that change the Phase 1 diagnosis itself. The timing of the cycle is not fixed: some actions can be reassessed within months, while others require multiple years before their effects can be fully evaluated.

The expected outcome of Phase 3 is a set of implemented actions with documented outcomes, a revised set of priorities and an updated roadmap cycle. With this, the firm re-enters Phase 1, closing the iterative loop on which the roadmap is built.

5.5 Roadmap use and facilitation

The roadmap is designed to be usable both by individual contractor firms and through facilitation by Skogsentreprenörerna, the Swedish contractor industry association. Individual

contractor firms operate with limited managerial capacity for sustained strategic work. If the roadmap is left as something each contractor must initiate alone, its practical use is limited to firms that already have substantial internal strategic capacity.

Skogsentreprenörerna occupies a structural position from which the roadmap can be made accessible to its member firms. The organization can provide shared analytical infrastructure including the reflection questions and templates set out in Appendix A, which are organized by the three aggregate dimensions, host facilitated workshops in which member firms work through the diagnostic and prioritization phases together, and aggregate the patterns that emerge across member firms, dimension by dimension, for dialogue with customers, policymakers and suppliers about the conditions that shape contractor work. Appendix A provides the operational material for both individual and facilitated use.

6. Discussion and Conclusion

This study set out to understand how ongoing transformations within the Swedish forestry industry impact forestry contractors and to develop a strategic roadmap that supports their adaptation. The empirical analysis identified three interacting aggregate dimensions of transformation pressure, and the roadmap structures contractor adaptation as an iterative movement from diagnosis through prioritization to action and reflection. This chapter interprets the findings in relation to the two research questions, develops the theoretical and practical contributions of the study, and addresses its limitations and directions for future research.

6.1 How ongoing transformations impact forestry contractors

RQ1 asked how ongoing transformations within the forestry industry impact Swedish forestry contractors. The empirical material shows that this impact is not best understood as a single disruptive change or as a question of whether contractors will continue to be needed. Contractors remain operationally central to the Swedish forestry value chain; the impact lies in the broadening and increasing complexity of the conditions under which contractor competitiveness has to be maintained.

Three interacting mechanisms shape this impact. Technological and digital change reaches contractors as a continuous reshaping of how work is planned, executed, recorded and reported, with digitalization in particular arriving as a fragmented set of customer-driven systems rather than the integrated digital value chain that Holmström (2020) describes as the sector's potential. Value-chain and business model transformation reaches contractors as changing conditions under which work is bought, coordinated, documented and valued, including asymmetric contract conditions, growing administrative burden and substantive sustainability and certification requirements. This extends Kronholm et al. (2021) and Eriksson and Sääf (2017), who documented contractor profitability and margin pressure but did not trace how multiple transformation pressures interact at the contractor level. Capability and professionalization pressures reach contractors as a structural operator shortage, broadening competence demands and a shift in what it takes to run a contractor firm well.

Critically, these mechanisms interact rather than act in isolation in the contractor's lived reality, as Section 4.1 set out: the same customer-driven digital system is at once a technological, an administrative and a competence pressure. This pattern of interacting and reinforcing pressures goes beyond what prior research has addressed. Näyhä and Pesonen (2014) described strategic change in the Nordic forest industry as multidimensional, and Onufrey and Bergek (2021) showed that transformation in mature industries is cumulative and path-dependent rather than abrupt. The present study confirms these characterizations and extends them by showing specifically how cumulative transformation pressures are filtered through the contractor's value-chain position, producing a qualitatively different experience of industry change than that of the larger actors who define many of the conditions under which contractors operate.

6.2 The strategic roadmap as a response to contractor transformation pressure

RQ2 asked how a strategic roadmap can be designed to support Swedish forestry contractors in adapting to these transformations. The roadmap developed in Chapter 5 structures adaptation as an iterative process in which the contractor moves between diagnosing exposure, prioritizing responses across cross-cutting content areas, acting on selected priorities and reflecting on outcomes to revise the next cycle. Its design draws on the strategic roadmapping tradition (Phaal, Farrukh & Probert, 2004) and on phase-based transformation frameworks that translate qualitative empirical material into structured, actionable outputs (Frishammar & Parida, 2018; Linde et al., 2021).

The central interpretive argument of the thesis is that the strategic question facing contractors is not whether to be reactive or proactive but where to be each. Forestry work involves substantial day-to-day reactivity to weather, site conditions, machine breakdowns and customer-driven planning changes, and this reactivity is part of the craft rather than a strategic deficit. The empirical material is clear, however, that reactivity also tends to dominate in strategic areas where deliberate preparation would be possible, such as the timing of machine investments, preparation ahead of contract renewals, and longer-term competence and recruitment planning. Drawing on Mintzberg and Waters (1985), proactive positioning as developed here is the creation of enough deliberate strategic clarity to support necessary operational reactivity, rather than the replacement of reactivity with predetermined plans. A

contractor with a structured diagnosis of its position is not freed from reacting to operational events, but is in a stronger position to interpret them strategically rather than only operationally.

This argument carries an important implication. The contractor's strategic position is shaped, enabled and constrained by the value chain within which the firm operates. The asymmetric pattern identified in Section 4.3.1 is interpreted here as a structural feature rather than a question of intent: the contractor's position, being capital-intensive, dependent on a small number of large buyers and exposed to friction in planning and contract conditions, produces a different lived experience of the same value chain than that of the actors who set many of its conditions. Contractors thus carry friction that larger value-chain actors do not always perceive, while the conditions under which their work is performed are substantially set elsewhere. Proactive positioning therefore cannot rely on external conditions improving on their own: where strategic agency exists for the contractor, it exists within the value-chain position rather than over it, which is why the roadmap is framed around proactive contractor positioning rather than correcting the value chain externally.

6.3 Theoretical contributions

This thesis makes two theoretical contributions. First, it contributes to the literature on industry transformation and forestry contractors by providing an integrated understanding of how ongoing transformations together impact Swedish forestry contractors. Prior research has addressed forestry transformation primarily at the level of the sector or bioeconomy (Näyhä & Pesonen, 2014), specific technologies such as digitalization (Holmström, 2020) or automation (Visser & Obi, 2021), or sustainability and certification dynamics (Lindahl et al., 2017). The more specific literature on Swedish forestry contractors has examined sector development (Häggström et al., 2013), business models and profitability (Kronholm et al., 2021) and economic performance (Eriksson & Sääf, 2017), but treats contractors either as one stakeholder group among several or as the subject of individual-transformation studies rather than as the focal actor facing multiple interacting pressures simultaneously. By identifying three aggregate dimensions and showing how they reinforce each other in the contractor's operating reality, this thesis moves the conversation from individual transformations analysed separately to an integrated, actor-level account of how industry transformation reaches and reshapes the contractor's strategic situation.

Second, this thesis contributes to the strategic roadmapping literature by extending roadmapping into the context of SME-scale, capital-intensive service providers operating under structural value-chain dependence. Strategic roadmapping has developed primarily in large industrial firms and whole-sector transitions (Phaal, Farrukh & Probert, 2004; Phaal & Muller, 2009; Phaal, 2015), and phase-based transformation frameworks have been applied to incumbent firms undergoing business model change (Frishammar & Parida, 2018) and to digital business model evaluation (Linde et al., 2021). These contributions have not addressed the specific situation of contractor-style actors who combine limited managerial capacity, high operational reactivity, capital-intensive assets and dependence on conditions set by larger value-chain actors. The roadmap developed here adapts the tradition to this context in three ways: by structuring adaptation as an iterative cycle that accommodates the ongoing and uneven nature of forestry transformation rather than assuming linear progression; by separating strategic-level guidance in the main text from operational templates in the appendix to keep the framework proportionate to resource-constrained firms; and by giving Skogsentreprenörerna a structural facilitator role that recognizes individual firms may lack the capacity to sustain strategic work alone.

6.4 Practical implications

The practical implications of the study extend to four groups of actors: forestry contractors, Skogsentreprenörerna, large forest companies and other customers, and technology suppliers and training actors. For forestry contractors, the central practical contribution is the roadmap itself, together with its operationalization in the appendix. It provides a structured way to evaluate exposure to ongoing transformations, prioritize across the content areas where strategic adaptation is most needed and revise priorities over time as conditions evolve. Contractors of different sizes, regions and service mixes can use the same roadmap structure and reach different concrete conclusions, which reflects the variation that characterizes the Swedish contractor sector.

For Skogsentreprenörerna, the roadmap offers both a facilitator framework for supporting member firms through workshops, shared analytical infrastructure and the reflection templates in the appendix, and a shared language for dialogue with forest companies, policymakers and

suppliers about the conditions that shape contractor work. For large forest companies and other customers, the findings clarify how planning quality, contract design, administrative demands, data systems and sustainability requirements filter down to the contractor level and affect contractor capacity. For technology suppliers and training actors, the implication is that technology adoption depends not only on technical functionality but on the economic, operational and competence conditions under which contractors can absorb new technology.

6.5 Limitations and future research

The study has several limitations that should be acknowledged. The research design is qualitative and contextually Swedish; findings provide depth and analytical relevance but do not claim statistical generalizability to other national forestry contexts. The study is future-oriented but not a forecast, and the roadmap is designed to remain useful under uncertainty rather than to assume that respondent expectations will prove accurate. The roadmap is empirically and theoretically grounded but has not been tested through actual contractor implementation; its usefulness in practice remains to be established through application. The collaboration with Skogsentreprenörerna strengthens contractor relevance but may have shaped which contractor concerns received particular attention, an effect partly balanced by including respondents from multiple actor categories across the value chain. The empirical material is also unevenly distributed: a minority of interviews were documented through structured notes rather than full transcripts and were used primarily for triangulation rather than direct quotation, and the group interview with three representatives of Skogsentreprenörerna is by far the longest transcript in the study. This gives the organization's account particular weight, a feature mitigated but not eliminated by triangulation across all eighteen interviews, as described in Section 3.2.2.

Several directions for future research follow. The roadmap could be tested in contractor practice by applying it in selected firms and following its use over time, with particular attention to how Skogsentreprenörerna can deliver the facilitator role outlined in Section 5.5. Comparative research across contractor segments (small versus large firms, harvesting versus silviculture, and different regions) could identify how the cross-cutting content areas play out across the sector. The contractor-customer relationship deserves dedicated study, particularly regarding contract design, planning quality, data access and risk-sharing. Technology adoption

under contractor constraints also warrants study, since it depends on investment economics, customer support, data ownership and operator competence as well as technical functionality. Finally, education, recruitment and competence development warrant focused research given the structural operator shortage and broadening competence demands documented here.

In conclusion, the future competitiveness of Swedish forestry contractors depends not on adapting to any single technology, regulation or contractual condition, but on developing a structured capacity to understand and respond to interacting transformation pressures over time. This thesis has offered forestry contractors and the actors that surround them a framework for that strategic preparedness, grounded in the empirical experiences of the Swedish forestry value chain and informed by the methodological tradition of strategic roadmapping.

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Appendices

Appendix A – Roadmap facilitation guide

This appendix operationalizes the three roadmap phases described in Chapter 5 through reflection questions and simple templates that contractors and Skogsentreprenörerna can use directly. The questions and templates are designed to remain proportionate to the resources available in SME-scale contractor firms; none of them require external consultancy support, specialist analytical tools or extended off-line time. They are intended to be useful both for individual contractor firms working through the roadmap internally and as the basis for facilitated workshops organized by Skogsentreprenörerna.

Phase 1: Evaluate current position
Reflection questions
The following questions guide a structured Phase 1 diagnosis. They are organized by the three aggregate dimensions identified in this study and can be worked through in a single session or across several shorter sessions.
On technological and digital transformation:

- Which customer-specific digital systems does the firm currently operate within, and how much administrative time do these systems consume together?
- Where would the firm be exposed if one more major customer system were added, or if an existing system substantially changed its requirements?
- What is the firm's current position on automation and operator-support technology, and which technology generation is the firm's current equipment fleet aligned with?
- How does the firm currently handle the data that flows through its operations, and what would change if data-ownership conditions tightened?

On value-chain and business model transformation:

- Which customers does the firm currently work with, and how dependent is firm utilization on the two or three largest customers?
- Which contract terms are most exposed to customer-driven change, and when does each major contract come up for renewal?
- How much of the firm's current administrative workload arises from documentation, reporting and certification requirements, and which of these are growing fastest?
- Which sustainability and certification expectations does the firm meet today, and which are likely to tighten in the next one to two years?

On capability and professionalization transformation:

- What is the firm's current operator base, and what is the expected operator turnover over the next two to three years?
- Which competences are most difficult to recruit or develop internally, and how is the firm currently addressing this?
- How is strategic and administrative work distributed within the firm today, and where does this distribution create bottlenecks?
- What would change for the firm if the structural operator shortage tightens further over the next three to five years?

Template: Current-position diagnosis

The Phase 1 deliverable is a structured diagnosis. A simple four-column template, completed for each of the three aggregate dimensions in turn, supports this:

Pressure	Current position of the firm	Exposure if the pressure tightens	Strategic gap or opportunity
-	-	-	-
-	-	-	-
-	-	-	-

Each row corresponds to one of the pressures the firm has identified through the reflection questions above. The final column distinguishes pressures that need attention in the next twelve to eighteen months from those that can be revisited in a later cycle.

Phase 2: Formulate strategic direction

Reflection questions

- Across the seven cross-cutting content areas (technology, competence, customer collaboration and contracts, operational efficiency, digital and data readiness, service scope, compliance), which two or three deserve deliberate strategic attention in the coming cycle, and why?
- Which content areas are explicitly being deprioritized in this cycle, with the understanding that they will be revisited later?
- Which actions belong to the longer-horizon category of service-scope, partnership or contract-design decisions, and should be treated as separate from shorter-horizon actions?
- What is the rationale behind each prioritized action, and what outcome is expected if the action succeeds?
- What assumption underlies the prioritization, and what would have to be true for the prioritization to remain valid through the cycle?

Template: Strategic action plan

The Phase 2 deliverable is a strategic action plan with prioritized actions, rationale and expected outcomes per action area. A simple template supports this:

Content area	Prioritized action	Rationale	Expected outcome	Horizon
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Phase 3: Implementation and reflection

Reflection questions

- Which actions from the Phase 2 plan have been piloted, on what scope and over what period?
- What were the intended outcomes recorded as rationale in Phase 2, and what actually happened?
- What unintended outcomes surfaced during implementation, and what do they reveal about the firm's capabilities or constraints?
- How did the action land with the relevant customers, partners or operators, and what does that reveal about the value-chain conditions the firm operates within?
- Which assumptions from the Phase 2 prioritization proved correct, which were partially correct, and which turned out to be wrong?
- Which priorities should be carried forward unchanged, which should be revised, and which should be retired entirely?
- Has the firm's exposure shifted in ways that change the Phase 1 diagnosis itself?

Template: Outcomes tracking and cycle update

The Phase 3 deliverable is a set of implemented actions with documented outcomes, revised priorities and an updated roadmap cycle. A simple template supports this:

Action	Pilot scope	Intended outcome	Actual outcome (intended + unintended)	Cycle update (carry, revise, retire)
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Notes for facilitated use

The reflection questions and templates above are written for direct use by contractor firms. They can also be adapted by Skogsentreprenörerna for use in facilitated workshops with member firms. Three modes of facilitated use are particularly suited to the structure: a regional Phase 1 workshop in which member firms work through the diagnostic questions and template in parallel and share patterns at the end; a thematic Phase 2 workshop organized around a specific content area (for example customer collaboration and contracts, or operator competence) in which firms produce their own action plans against a shared template; and a Phase 3 reflection session in which member firms exchange what their pilots revealed before entering the next cycle. In each case, the firm produces its own outputs while the workshop structure reduces the strategic burden on individual contractors and lets the organization aggregate patterns across the membership.

Appendix B – Semi-structured interview guide

The guide is reproduced in the original language in which the interviews were conducted

Intervjuguiden
Syfte med intervjuerna:
Syftet är att identifiera och analysera strukturella förändringsdrivkrafter inom den svenska skogsindustrin under en cirka 15-årig tidshorisont, med särskilda analytiska nedslag mot 2030, 2035 och 2040. Intervjuerna syftar till att förstå hur dessa förändringar kan påverka värdekedjans logik, maktförhållanden och handlingsutrymme särskilt för skogsentreprenörer samt vilka strategiska vägval som kan bli relevanta.
Övergripande intervjustruktur
1. Nulägesförståelse
2. Strukturella förändringsdrivkrafter (teknik, policy, marknad, organisation)
3. Tidsdimension: Hur kan detta utvecklas mot 2030, 2035 och 2040?
4. Konsekvenser för värdekedjan och maktförhållanden
5. Strategiska implikationer för skogsentreprenörer
Gemensamma kärnfrågor (alla aktörer)
• Vilka strukturella förändringar bedömer ni som mest betydelsefulla för skogsindustrin på 10–15 års sikt?
• Vilka drivkrafter ligger bakom dessa förändringar (teknik, policy, marknad, samhällskrav)?
• Hur tror ni dessa drivkrafter kan utvecklas fram till 2030? Och därefter mot 2035 och 2040?
• Hur kan dessa förändringar påverka relationer och maktförhållanden i värdekedjan?
• Vilka aktörer tror ni stärker respektive försvagar sin position över tid och varför?
Skogsbolag och skogsägare
• Hur ser ni att entreprenörsrollen kan förändras på 10–15 års sikt?
• Kommer kravbilden på entreprenörer att förändras? Hur?
• Hur ser ni på framtida kontraktsformer och riskfördelning?
• Vilka kompetenser eller kapabiliteter blir viktigare framöver?
• Finns det strukturella hinder idag som begränsar entreprenörernas strategiska utveckling?
Skogsentreprenörer
• Vilka förändringar upplever ni redan idag påverkar er affär mest?
• Vilka förändringar tror ni kommer påverka er mest fram till 2030? Och på längre sikt?
• Hur upplever ni er position i värdekedjan – vilka är era största beroenden?
• Vilka strategiska vägval ser ni som möjliga för att stärka er långsiktiga position?
• Vad skulle krävas för att ni ska kunna arbeta mer proaktivt snarare än reaktivt?
• Finns det något kopplat till skogsentreprenörer eller deras framtida roll som du tycker diskuteras för lite idag?
Maskintillverkare och teknikleverantörer
• Vilka teknologiska språng bedömer ni som mest sannolika inom 10–15 år?
• Hur påverkar elektrifiering, digitalisering och AI er långsiktiga strategi?
• Hur ser ni på entreprenörernas roll i utveckling och implementering av ny teknik?
• Vilka hinder finns för bred implementering av ny teknik i branschen?

Forskning, innovation och branschorganisationer
• Vilka långsiktiga transformationstrender bedömer ni som mest avgörande?
• Hur fungerar innovationssystemet i skogsindustrin idag?
• Vad hindrar mindre aktörer från att ta en mer aktiv roll i utvecklingen?
• Vilka frågor bör få större strategiskt fokus i branschdialogen?
Avslutande reflektionsfråga
• Om ni blickar mot 2040 – vad tror ni kommer vara den största strukturella skillnaden jämfört med idag, och hur bör skogsentreprenörer ha positionerat sig för att vara väl rustade då?

Appendix C – Data structure shown as a table

Data structure: progression from 1st-order observations to three aggregate dimensions

1st-order observations	2nd-order themes	Aggregate dimensions
AI-supported machine operation and decision support to the operator [R8, R9]	**Digitalization and data flows**	**1. Technological and digital transformation**
Drone and satellite forest mapping at tree-level resolution [R8]		
Contractors juggling multiple uncoordinated customer portals; customer-driven systems often add to admin [R10, R14]		
Industry-wide perception that forestry is behind on digitalization [R14, R1]		
Automation expected stepwise, not as a single jump [R7, R8, R9]	**Stepwise automation and mechanization**	
Remote control and one-operator-multiple-machines as the near-term productivity model [R7, R8]		
Mechanized planting growing under cost and speed pressure [R14, R18]		
Forestry among the last sectors expected to see full unmanned operations [R1, R4]		
Three machine-maker tech levers: digitalization, automation, electrification [R9]	**Machine evolution**	

1st-order observations	2nd-order themes	Aggregate dimensions
Electrification not near-term in Swedish forestry; Scandinavia likely first when it arrives [R4, R9]		
Lighter, more terrain-friendly machines expected over time [R4, R9]		
Asymmetric tension: contractors perceive friction, large companies do not [R5, R6, R10, R14, R15, R16, R11–R13]	**Contractor–customer value-chain dynamics**	**2. Value-chain and business model transformation**
Oligopoly customer concentration (seven to eight large buyers) [R14]		
Risk concentration on the contractor: "can't be the buffer for everyone else's uncertainty" [R10]		
Index clauses for fuel, interest, parts and downtime becoming critical to contracting [R4, R10, R15, R16]		
Global competition (China, South America) driving downstream cost-down pressure onto contractors [R5, R17]		
Servitization and total-cost-of-ownership discussions among machine manufacturers [R9, R18]	**Emerging business models**	
Contractors prefer to own their capital; resistant to leasing or mobility-as-service models [R18]		
Service-integrator role emerging at larger scale (e.g. managing 100+ contractors) [R17]		
Administrative load on contractors growing year over year [R10, R11–R13, R15, R16]	**Regulatory and administrative burden**	
Environmental NGO pressure cascading through customers to contractor operations [R1, R6]		
Knowledge requirements (new directives, training, documentation) growing [R11–R13]		
Carbon-neutral 2045 with interim 2030 milestones; cascading reporting demands on contractors [R4, R11–R13]	**Sustainability and certification requirements**	

1st-order observations	2nd-order themes	Aggregate dimensions
Tree species diversification away from monocultures shifting contractor work mix [R17]		
Biodiversity preservation versus timber production tension shaping operating conditions [R3, R6]		
FSC and PEFC certification as baseline expectation; non-negotiable for contractor work [R6]		
Operator shortage acute across the industry [R6, R10, R14, R15, R16]	**Operator shortage**	**3. Capability and professionalization transformation**
Aging workforce and pension departures [R6]		
Large companies running own training programs (e.g. SCA: 48 trainees/year) [R6]		
Operators need broader skills: technical, judgment, responsibility, digital systems [R4, R10, R15, R16]	**Rising competence demands**	
Future contractor combines machine, digital, quality and business competence [R4]		
Industry attractiveness tied to profitability [R11–R13]		
Larger contractors increasingly need professional management, not only administrative support [R15]	**Strategic professionalization**	
Generational succession as cultural feature of the contractor sector [R18]		
Long-term contracts become more important as machines and digital systems become more expensive [R4]		